

DEVELOPMENT FOR ARTIFICIAL INTELLIGENCE

COURSE ASSESSMENT

CivicRoute AI Supporting Data

Data Dictionary and Category Guide

The supplied structured data is synthetic educational materials. They do not represent real residents, real municipal requests or real personal information.

Assessment scope

Only the supplied structured CivicRoute AI dataset, namely `civic_requests_prepared.csv` is required for the assessment. The TensorFlow/Keras starter notebook is also supplied to aid your coding efforts.

Request Categories

Category label	Plain-language meaning
pothole	Damage or a hole in a road surface.
water_leak	Visible or reported water escaping from municipal infrastructure.
broken_streetlight	A streetlight that is not operating correctly.
illegal_dumping	Waste or unwanted material left in an unauthorised public area.

Assessment Dataset Fields

The following fields appear in the `civic_requests_prepared.csv` dataset file.

Field	Type	Description	Notes
request_id	text	Unique educational request identifier.	Identifier; exclude from model inputs.
created_hour	integer	Hour of day when the request was created, from 0 to 23.	Prepared numerical field; possible model input.
neighbourhood	category	Neighbourhood linked to the request.	Categorical field; consider fairness and reporting-pattern limitations.
channel	category	Submission channel: mobile app, web portal, call centre or service office.	Categorical field with 4 missing values to inspect and handle.
urgency_level	category	Prepared urgency indication: low, medium or high.	Prepared categorical field; input feature, not a final routing decision.
recent_rain_flag	category	Whether recent rain was recorded.	Yes/no categorical field.
night_report_flag	category	Whether the request was submitted at night.	Derived from <code>created_hour</code> ; consider whether both fields are needed.
repeat_report_count	integer	Number of similar nearby reports recorded recently.	Prepared numerical feature.
image_quality_score	decimal	Prepared image-quality score from 0 to 1.	Prepared image-quality metadata score from 0 to 1; 10 values are missing. No image files are supplied.
nearby_asset_age_years	decimal	Approximate age of the nearby infrastructure asset.	Prepared numerical feature.
description_word_count	integer	Number of words in the prepared request description.	Prepared numerical feature; the description text itself is not supplied.
previous_resolution_days	decimal	Days previously required to resolve a similar case.	Prepared numerical feature with 9 missing values to inspect and handle.
category_label	category	Target request category.	Prediction target with four balanced request categories.

Dataset Summary

Item	Summary
Assessment file and size	civic_requests_prepared.csv: 600 records and 13 fields.
Target categories	4 balanced categories with 150 records each: pothole, water_leak, broken_streetlight and illegal_dumping.
Missing values	channel: 4; image_quality_score: 10; previous_resolution_days: 9. All other fields are complete.
Assessment use	Inspect and slice the data, handle the small preparation issues, encode or transform selected fields, then build one small TensorFlow/Keras classifier.